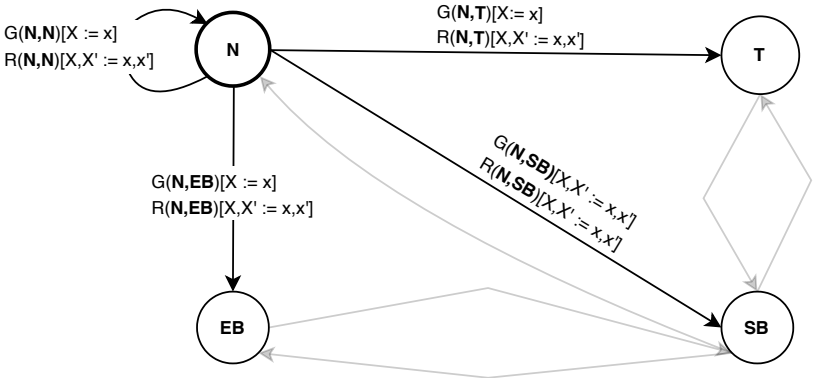




Guard and Reset

$G(\mathbf{N}, \mathbf{N})[X := x] = t = D_FREQ$

$G(\mathbf{N}, \mathbf{T})[X := x] = t < D_FREQ \wedge sc < STALE \wedge sr_dist \wedge \neg rc_dist \wedge \neg c_dist$

$G(\mathbf{N}, \mathbf{SB})[X := x] = t < D_FREQ \wedge sc < STALE \wedge rc_dist \wedge \neg c_dist$

$G(\mathbf{N}, \mathbf{EB})[X := x] = t < D_FREQ \wedge sc < STALE \wedge \neg c_dist$

$R(\mathbf{N}, \mathbf{N})[X, X' := x, x'] = update[X, X' := x, x']$

$R(\mathbf{N}, \mathbf{T})[X, X' := x, x'] = R(\mathbf{N}, \mathbf{SB})[X, X' := x, x'] = R(\mathbf{N}, \mathbf{EB})[X, X' := x, x'] = staleness_reset[X, X' := x, x']$